



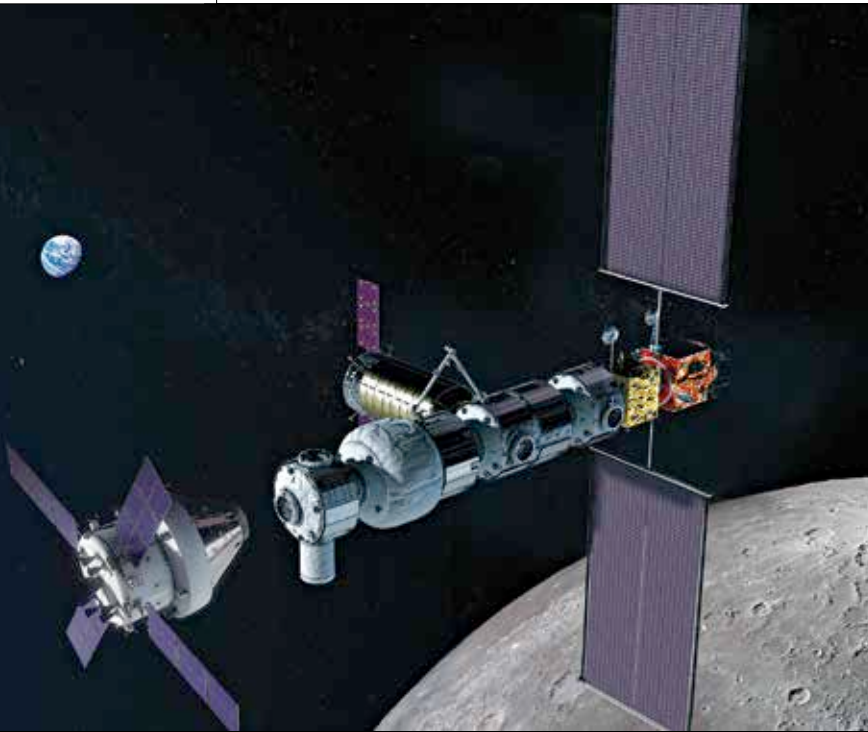
### Criticism of the gateway

A panel of experts outline why a space station in orbit around the Moon may be an unnecessary idea. <https://digit.in/may20-71>



### Gateway no longer critical

NASA removed the Lunar Gateway from its critical roadmap for the Artemis missions to the Moon. <https://digit.in/may20-72>



# LUNAR GATEWAY

For future pioneering missions to Mars and beyond, the stepping stone is a space station that acts as a staging area, lab and office, deployed in orbit around the Moon.

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ASA is building a space station in orbit around the Moon, which will be used as an office

space for astronauts on the crewed Artemis missions to the Moon. It will also be used as a laboratory for science experiments, and the staging area for demonstrating new technologies for deep space missions to the outer Solar System,

including Mars. Unlike the International Space Station, the Gateway will not be occupied throughout the year, but will only have visiting staff for a few months every year. The rest of the year, the Gateway will run autonomously, and will continue to conduct experiments even when not occupied by humans. For both manned, unmanned and autonomous probes, the Gateway can act like an airport, allowing the spacecrafts to dock, refill the propellant tanks, and be serviced. NASA is not working on the Gateway on its own,

there is support from other national space agencies including JAXA, ESA and CNSA. Apart from that, a whole bunch of private companies including Lockheed-Martin, Boeing, SpaceX, Northrop Grumman, Sierra Nevada Corporation, and Bigelow Aerospace are supporting the efforts. In fact, the planning for the spaceship is so detailed, that NASA has taken inputs from all the private spaceflight companies mentioned above, after asking them to conceptualise and design their own versions of the Gateway, each of which could theoretically actually be built and deployed.

While the ISS was expanded over a series of 34 launches, the Gateway is planned to be assembled over merely six launches. The brand new Space Launch System will deploy the main components into Earth's orbit, and the Orion spacecraft will then ferry them to orbit around the Moon. The journey from the Earth to the Moon takes around five days. Like the ISS, the Gateway will have a modular design too, with components that fit into one another. Visiting spacecraft can dock, and there will be airlocks for extravehicular activities for the astronauts. While the ISS had a living space equivalent to a six room house, the Gateway has a habitat that is comparable to a studio apartment.

The entire Gateway will be assembled in the target orbit itself, and can be thought of as a chain of modules. The first module, scheduled for a 2022 launch, is to provide power and propulsion to the entire platform. It will be used to make sure that the Gateway stays in the required orbit, and will also host the communications systems. The Esprit module will have an airlock, and is dedicated to science. It hosts various experiments for institutions and bodies around the world, and a portion of the space is allocated to storing extra propellant, as well as communication systems. The Esprit module is scheduled for a 2023 launch. Next is